Alt_MSP430 Business Case

The MSP430 microprocessor from Texas Instruments is low power and has very good code density. However it now suffers from limited address space. The original ISA also omits one very useful addressing mode: register indirect with pre-decrement.

The solution herein increases the set of data sizes from two (byte and 16-bit) to three (byte, 16-bit and 24 or 32-bit). The third data size supports larger addresses using a register file of 24 or 32-bit registers. By supporting the third data size larger addresses have the support of the full ISA. The third data size comes at a cost: some op-codes are only available through use of a prefix instruction.

The second fix is to support the five addressing modes: register, register indirect, register indirect post increment, register indirect pre decrement and register indirect with immediate offset. The five register modes are accomplished by limiting access to the register file to twelve general purpose registers. Five modes times twelve registers is sixty codes fitting into six bits. Four codes remain which can be used for immediate, absolute, frame relative, and PC relative. One of the twelve registers is the stack pointer.

The MSP430 ISA has a memory to memory mode. The author prefers replace operations which allow all four register indirect modes in specifying the destination address for the result.

Those instructions using an immediate constant or offset have the immediate length specified in the immediate. As a result 8 and 16-bit displacements or constants are fetched along with the 16-bit instruction (the memory interface is arranged so that 32-bit non-aligned memory reads take a single memory cycle).

The spirit of the MSP430 is retained and address space matches modern needs. A full set of addressing modes is provided. Half-word alignment of instructions is lost. Some op-codes require a prefix instruction. Prefix instructions can also be used to expand the register file along with additional op-codes and a third register field.